STANBUL SABAHATTIN ZAIM UNIVERSITY Faculty of Engineering and Natural Sciences

BIM432 NATURAL LANGUAGE PROCESSING SEMESTER PROJECT REPORT

PROJECT TITLE: AI vs Human Text Classification: A Machine Learning Approach to Detect AI-Generated Text

GitHub Repository: https://github.com/fzeynepgoren/nlp-ai-text-classifier.git

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TABLE OF CONTENTS

1. Introduction

In recent years, rapid technological developments in Natural Language Processing (NLP) and Deep Learning have contributed to the widespread integration of Large Language Models (LLMs) into many areas of daily life. Generative artificial intelligence systems such as OpenAI’s GPT series, Anthropic’s Claude, and Google’s Gemini have become highly capable of analyzing and reproducing patterns in human language.

These models provide significant practical benefits in tasks such as text summarization, code generation, translation, and creative content production, leading to important changes in the digital environment. However, these developments have also introduced important ethical and security-related concerns. As AI-generated text becomes increasingly natural and fluent, distinguishing it from human-written text has become more difficult. This situation raises concerns in areas such as academic integrity, journalism ethics, cybersecurity, and misinformation. In particular, the use of AI tools by students to generate assignments or essays, as well as the rapid spread of fake news and synthetic content through digital platforms, has increased the need to question the source and originality of written content.

The main objective of this project is to develop a reliable binary classification pipeline that can distinguish between human-written and AI-generated texts with high accuracy. For this purpose, the project uses the AI vs Human Text Dataset and applies text preprocessing and feature extraction techniques based on TF-IDF (Term Frequency-Inverse Document Frequency). In the preprocessing stage, the text is cleaned and tokenized, while TF-IDF is used to represent the relative importance of terms within the corpus.For model training, two machine learning algorithms are used: Logistic Regression, which provides a traditional and relatively interpretable baseline, and Support Vector Machines (SVM), which are commonly effective in high-dimensional feature spaces. Another focus of the study is to compare how different preprocessing and feature extraction choices affect the performance of these two models using multiple evaluation metrics.

2. Related Work

In the Natural Language Processing literature, AI text detection has developed alongside generative language models and is often discussed as an ongoing competition between text generation and text detection methods. With the introduction of Transformer architectures, language models became increasingly effective at modeling semantic and syntactic patterns in human language. As a result, traditional text classification and plagiarism detection approaches have faced new limitations when applied to AI-generated content.

Early detection methods in the literature generally focused on identifying statistical irregularities in generated text. For example, tools such as GLTR (Giant Language Model Test Room) analyzed word probability distributions and the predictability of language model outputs (Gehrmann et al., 2019). In this context, two commonly discussed indicators have been perplexity and burstiness. Perplexity measures how surprising or unpredictable a text is for a language model, while burstiness refers to variation in sentence lengths and syntactic structures within a text. Approaches such as DetectGPT proposed by Mitchell et al. (2023) also use zero-shot detection strategies based on model behavior and probability patterns. However, more recent generative AI systems are increasingly able to produce texts with greater stylistic variation, which makes purely statistical detection methods less reliable in some cases.

As the limitations of statistical approaches became more visible, researchers also explored deep learning-based architectures such as BERT (Bidirectional Encoder Representations from Transformers), RoBERTa-based detectors, and other Transformer-based classifiers. For example, OpenAI's RoBERTa-based detector and related Transformer-based classification studies have reported strong performance in detecting machine-generated text (Solaiman et al., 2019; Ippolito et al., 2020). Nevertheless, these models often require greater computational resources and may be harder to interpret due to their black-box nature. This can be a disadvantage in settings where transparency and explainability are important.Recent studies also indicate that traditional machine learning approaches can still be competitive, especially when the goal is to identify lexical and stylistic patterns in generated text. In particular, high-dimensional lexical representations based on n-grams and TF-IDF can be effectively combined with linear models such as Logistic Regression and Support Vector Machines (SVM). These approaches may offer a practical balance between performance, computational efficiency, and interpretability. Therefore, the methodology used in this project is based on a classical NLP pipeline that aims to provide strong classification performance while remaining relatively simple and explainable.

3. Dataset Description

In this study, a text dataset named AI_Human.csv was used to distinguish between human-written and AI-generated texts. The dataset mainly consists of essays, articles, and informative texts, providing variation in text length and writing style. This diversity supports the model's ability to learn from different types of written content and improves the reliability of the evaluation process.

3.1. Dataset Statistics and Class Distribution

The raw dataset used in this study contains a total of 487,235 text samples. Structurally, the dataset consists of two main columns:

text: Contains the raw text data to be analyzed.

generated: Represents the binary target label indicating the source of the text. Human-written texts are labeled as 0.0, while AI-generated texts are labeled as 1.0.

When examined in terms of class distribution, the dataset is not perfectly balanced, but it has a reasonably acceptable distribution for model training and evaluation. The dataset was split into 80% training data and 20% test data. The test set consists of 97,447 samples, with the following class distribution:

Human-written text (Label 0.0): 61,112 samples, approximately 62.7%

AI-generated text (Label 1.0): 36,335 samples, approximately 37.3%

3.2. Text Characteristics and Challenges

The human-written texts in the dataset, such as student essays and articles on topics like "Electoral College" and "Car Usage," generally contain natural language variation, spelling inconsistencies, and differences in sentence structure. These characteristics reflect the variability commonly observed in human writing. In contrast, AI-generated texts, labeled as 1.0, tend to display more consistent grammar and a more regular writing structure. However, they may also include repeated word patterns and a certain level of structural uniformity. These differences create useful stylistic signals that machine learning models can learn during classification. Considering the size of the dataset, approximately 1.1 GB, and the variety of vocabulary it contains, the data provides a substantial basis for learning lexical and stylistic patterns. However, due to the structural complexity and raw format of the texts, it is necessary to apply Natural Language Processing preprocessing steps and feature extraction methods such as TF-IDF before training machine learning models such as Logistic Regression and Support Vector Machines

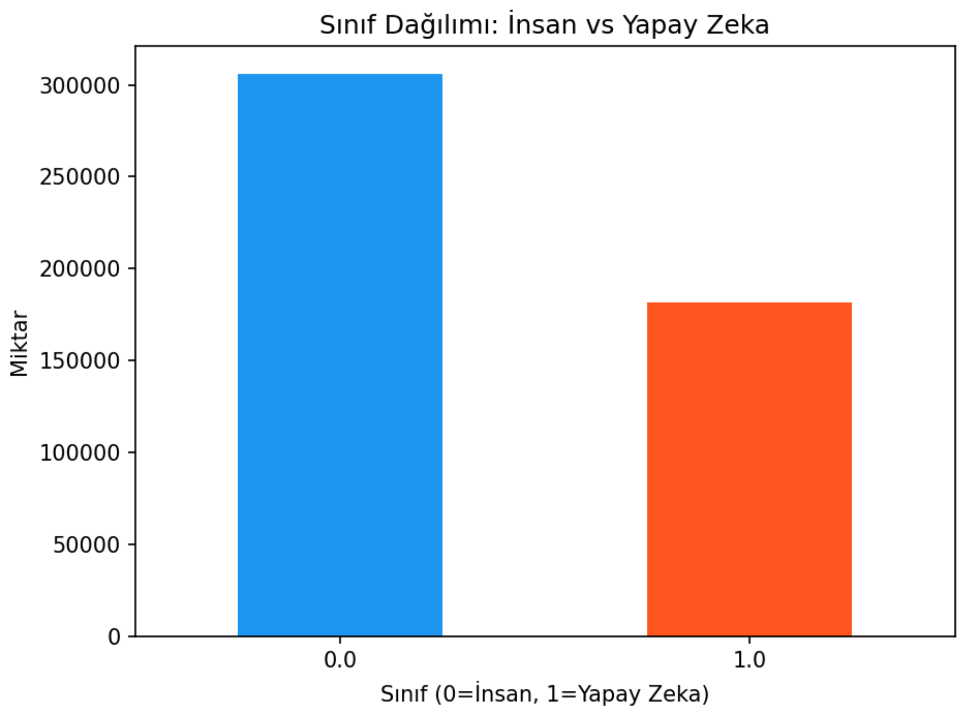


Figure 1. Comparison of Model Accuracies

4. Methodology

This project's classification infrastructure is based on classical and robust Natural Language Processing (NLP) architectures that transform textual data into numerical feature spaces and feed them into machine learning algorithms. The developed pipeline consists of three main stages: Data preprocessing, feature extraction, and model training.

4.1. Data Preprocessing

In order to reduce the noise contained in raw text data and ensure that the models focus only on semantic patterns, standard preprocessing steps were applied using the nltk library in Python. The implemented steps and their justifications are as follows:

Lowercasing:
All texts were converted to lowercase using the lower() function in order to prevent the classifier from perceiving the same word ("Apple" and "apple") as different features.

Punctuation Removal:
Symbols that do not carry semantic value and unnecessarily expand the vector space were cleaned using Python's string.punctuation module.

Stopword Removal:
Conjunctions and prepositions such as "the", "is", and "at", which occur very frequently in language but do not contribute significantly to the actual context of the text, were removed in order to reduce computational cost and increase the weight of informative words.

4.2. Feature Extraction

Since machine learning algorithms cannot process text directly, textual data must be transformed into numerical vectors. In this study, instead of the simple Bag of Words method, the TF-IDF (Term Frequency – Inverse Document Frequency) algorithm was preferred, as it statistically scales the importance of words within documents.The TF-IDF weight of a term "t" in a document "d" within a corpus "D" consisting of "N" documents was calculated according to the following logic:

Term Frequency (TF):
Measures how frequently a word appears in the relevant document.

TF = Number of occurrences of the related word in the document / Total number of words in the document

Inverse Document Frequency (IDF):
Measures how rare (and therefore distinctive) the word is across the entire dataset.

IDF = log(Total number of documents / Number of documents containing the word)

Final feature vector: TF-IDF = TF × IDF

Using the TfidfVectorizer from the scikit-learn library integrated into the code, the matrix size was limited to a maximum of 5,000 most important features (max_features=5000) in order to optimize computational cost and prevent overfitting, resulting in a sparse matrix representation.

4.3. Model Training and Development

To perform classification within the generated 5000-dimensional feature space, the dataset was divided into 80% training and 20% testing sets using the train_test_split function (random_state=42) for reliability and reproducibility. Two different machine learning algorithms were trained for classification:

Logistic Regression:
At the initial stage, Logistic Regression was preferred as a baseline model due to its high interpretability and controllability. As a linear model, Logistic Regression calculates the probability that a text was generated by artificial intelligence using the sigmoid function. To ensure proper optimization convergence, the iteration number was set to 1000 (max_iter=1000).

Support Vector Machines (SVM):
In the second stage, LinearSVC was employed, as it mathematically performs best at drawing the optimal separating hyperplane between two classes (Human vs AI) and maximizing the margin within the high-dimensional sparse feature spaces generated by TF-IDF. To ensure model convergence, the iteration number was increased to 2000 (max_iter=2000).

5. Experiments

The Logistic Regression and Support Vector Machine (SVM) models developed within the scope of the project were subjected to a comprehensive evaluation on the test dataset. Beyond maximizing classification performance, a controlled experiment (Ablation Study) was designed in this study in order to examine the structural effects of data preprocessing steps on the predictive power of the model. One of the main experiments conducted in this study was the comparison of model performance between preprocessed and raw text data. This comparative analysis on the Logistic Regression model revealed an unexpected result that differs from common assumptions in machine learning literature and provides insight into the characteristics of AI-generated texts.

Accuracy Rate of the Preprocessed Model: 98.90%
Accuracy Rate of the Raw Text Model: 99.10%

Contrary to the traditional Natural Language Processing approach, avoiding data cleaning (preserving punctuation marks, uppercase/lowercase distinctions, and stopwords) increased the model’s accuracy instead of reducing it. The reason for this phenomenon is based on a fundamental difference between the text generation habits of human writers and artificial intelligence systems. Large Language Models (LLMs) generally apply perfectly flawless spelling, grammar, and punctuation rules. The fact that artificial intelligence places commas, periods, and quotation marks with mathematical precision effectively served as a massive “hidden feature” for our detection model. Humans, on the other hand, tend to make typing mistakes and create irregular structures while writing. Removing punctuation marks during preprocessing eliminated the “perfect grammar” signature that significantly helped the model identify AI-generated content, thereby making detection slightly more difficult.

6. Results

The performance of the Logistic Regression and Support Vector Machine (SVM) models trained through the steps described in Section 4 was evaluated on the test dataset. The evaluation was conducted on a previously unseen test set containing 97,447 text samples, including 61,112 human-written texts and 36,335 AI-generated texts.

6.1. Model Performances

To evaluate the models more comprehensively, the analysis was not limited to Accuracy alone. Precision, Recall, and F1-Score were also calculated in order to assess the models from different performance perspectives. Both models achieved strong results on the test set:

Model	Sınıf	Precision	Recall
Lojistik Regresyon	İnsan (0.0)	0.99	1.00
	Yapay Zeka (1.0)	0.99	0.98
LinearSVC (SVM)	İnsan (0.0)	1.00	1.00
	Yapay Zeka (1.0)	1.00	0.99

When the results are examined, the SVM model achieved slightly better performance than the Logistic Regression model. In the initial evaluation, SVM reached an accuracy of 99.72%, while Logistic Regression reached 99.14%. This difference can be explained by the fact that SVM is often well-suited for high-dimensional and sparse TF-IDF feature spaces, where it can learn an effective separating hyperplane between classes.

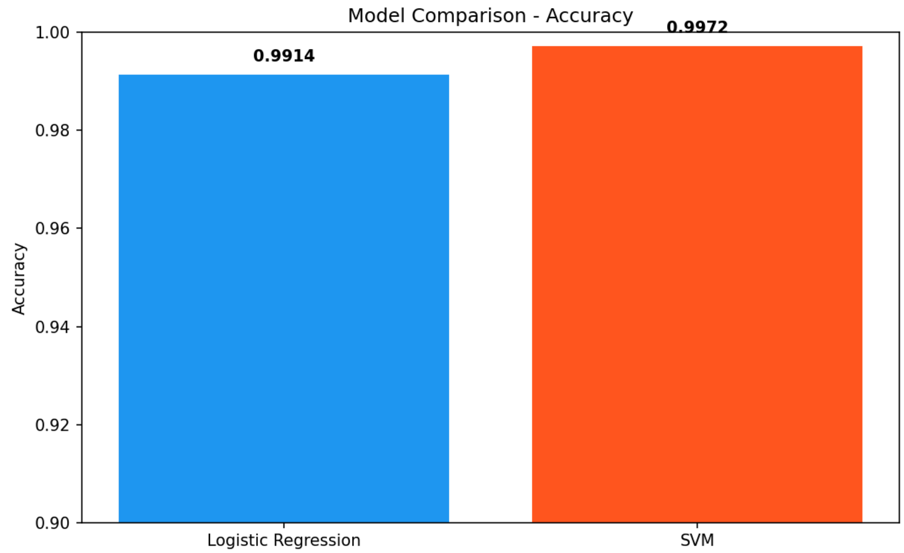
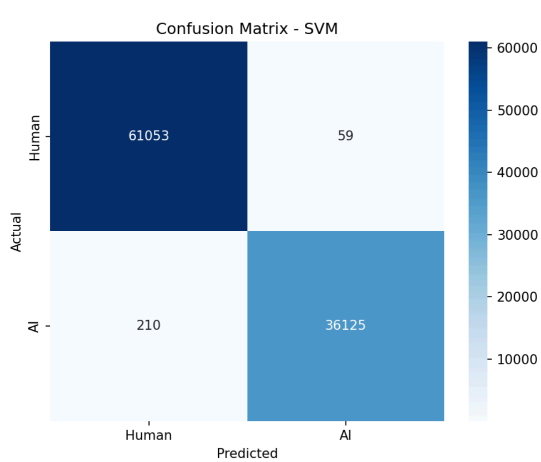


Figure 2. Confusion Matrix of Logistic Regression

6.2. Confusion Matrix Analysis

To examine the types of errors made by the models, especially False Positives and False Negatives, confusion matrices were analyzed.

- **Logistic Regression Errors:** The model made 840 errors out of 97,447 test samples. It incorrectly classified 267 human-written texts as AI-generated (False Positive) and classified 573 AI-generated texts as human-written (False Negative).
- **SVM Errors:** the total number of errors was lower, with 269 errors in total. The model incorrectly classified 59 human-written texts as AI-generated and missed 210 AI-generated texts, labeling them as human-written



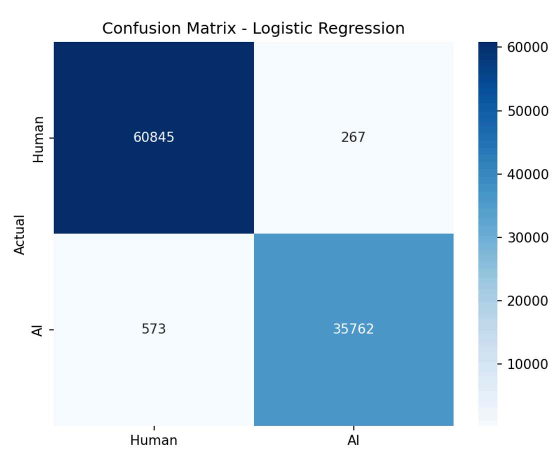


Figure 3. Confusion Matrix of LinearSVC (SVM)

In both models, False Negatives, where AI-generated texts are classified as human-written, occurred more frequently than False Positives. This observation is consistent with the discussion in Section 2, where recent language models are described as increasingly capable of imitating certain aspects of human writing style, such as variation in structure and expression. However, this result should be interpreted as supporting evidence rather than definitive proof.

Overall, the results suggest that traditional machine learning models combined with TF-IDF-based feature extraction can still achieve strong performance in AI text detection tasks. In this study, the SVM model reached an accuracy close to 99.7%, showing that a relatively simple and interpretable pipeline can be highly effective for this dataset.

7. Discussion

This study shows that traditional Natural Language Processing (NLP) methods can be highly effective and computationally efficient for the task of AI text detection, even without relying on more resource-intensive Deep Learning architectures. The findings can be discussed from both technical and conceptual perspectives under four main points:

7.1. Interpretability and the Power of Traditional NLP

The black-box nature of deep learning models, which was discussed as a limitation in Section 2, was addressed in this project by using more interpretable traditional machine learning models such as Logistic Regression and SVM. In addition to achieving strong accuracy, these models also provided an important advantage in terms of explainability. Through feature importance analysis, it was possible to examine which terms contributed most strongly to the models' decisions. For example, words such as "Additionally" ,"Conclusion" and "Essential" appeared as indicators associated with more formal and template-like AI-generated writing. On the other hand, words such as "going" ,"would" and "almost" were associated with more variable and less rigid human-written texts. These findings helped make the classification process more transparent and easier to interpret.

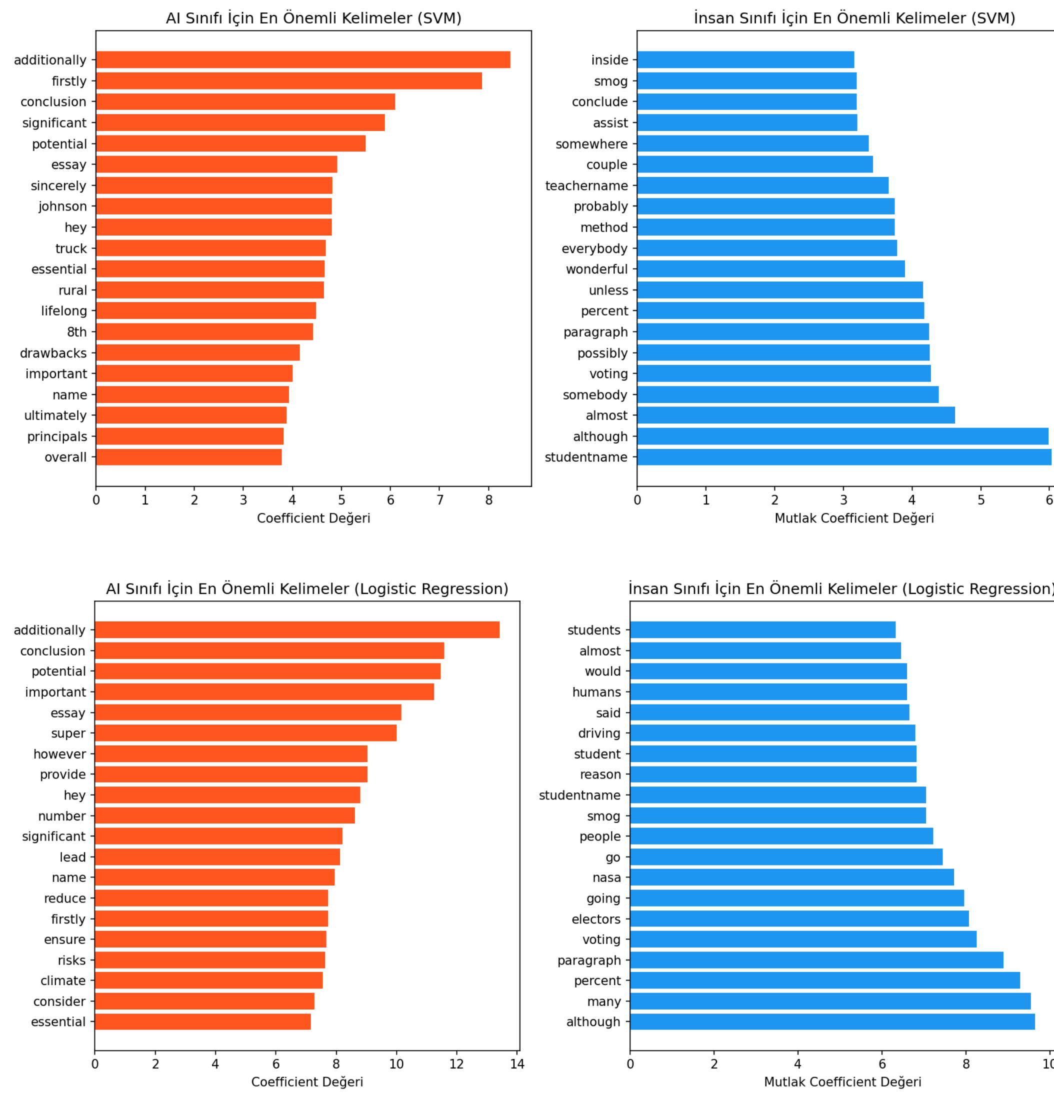


Figure 4. Feature Importance (LinearSVC)

7.2. The "Perfect Grammar" Vulnerability of LLMs

The preprocessing comparison discussed in Section 5 suggests that some surface-level writing patterns can be useful for AI text detection. In particular, AI-generated texts often show more consistent spelling, punctuation, and syntactic structure, while human-written texts may include more variation, informal usage, or minor typographical inconsistencies. In this study, preserving such features in the raw text setting helped the model capture stylistic differences between human-written and AI-generated texts.

7.3. Error Analysis and Stylometric Overlap

When the errors made by the models in the confusion matrices (False Positives and False Negatives) were examined, the gray areas between human and machine language were clearly mapped:

False Positive Situation (Classifying Human as AI):
When the human-written texts misclassified by the model were analyzed, they were found to consist of overly standardized job application letters such as “Hello, my name is Generic_Name... Today I am here because...” or exam essays written with rigid templates such as “I agree because some students are not as active as others...”. Excessively formal (robotic) human writing misled the model.

False Negative Situation (Classifying AI as Human):
The AI-generated texts that the model failed to detect were identified as texts intentionally containing subjectivity (first-person narration), empathy, and role-playing elements through prompt engineering, such as “My Dear Citizens...” or “I believe that the school administration...”.

7.4. Limitations and Future Work

The high accuracy rate of 99.72% may stem from the fact that the texts in the dataset possessed a specific informative structure (essay/article format). One of the fundamental limitations of this project is that the cross-domain performance of the model on different text types (such as Twitter posts, WhatsApp messages, or lines of code) has not been tested. In future studies, beyond word-based vectorization approaches such as TF-IDF, it may be possible to develop hybrid detection mechanisms that combine semantic pauses within sentences, sentiment volatility, and purely stylometric features independent of word-based representations.

8. Conclusion

This project presents an effective, computationally efficient, and explainable approach to the problem of AI text detection, which has become an important topic in Natural Language Processing (NLP). Using a dataset of **487,235 text samples**, the study shows that strong results can be achieved with traditional machine learning models without relying on deep learning architectures or intensive GPU-based computation.

Using n-gram and TF-IDF-based lexical feature extraction, the Logistic Regression and Support Vector Machine (SVM) models achieved high classification performance. In the initial evaluation, Logistic Regression reached **99.14% accuracy**, while SVM reached **99.72% accuracy**. These results indicate that lexical and stylistic patterns in AI-generated text can be effectively captured by classical machine learning methods.

One important finding of the study is that the grammatical consistency and structured writing patterns commonly observed in AI-generated texts can serve as useful signals for classification. Rather than treating these characteristics as hidden or perfectly reliable indicators, the results suggest that they provide meaningful statistical patterns that help distinguish AI-generated text from human-written text within this dataset.

Overall, this study demonstrates that classical NLP pipelines can still offer a strong balance between high accuracy, low computational cost, and explainability. Through feature importance analysis, the models also provide insight into which words and stylistic patterns influence classification decisions, making the approach more transparent than many black-box deep learning alternatives.

9. References

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